



Research Article

Trusted wills for digital assets using blockchain: a practical case

Jesus Hernando-Corrochano^{a, *}, Rafael Pastor-Vargas^{b, *}, Roberto Hernández-Berlinches^b^a Computer Science Department, UC3M, Leganés 28913, Spain^b Communications and Control Systems Department, ETSII, UNED, Madrid 28040, Spain

ARTICLE INFO

Keywords:

Digital assets
Blockchain systems
Digital legacy
Smart contracts

ABSTRACT

The scope of this work focuses on the digital assets of citizens and their personal and private inheritance. The aim is to protect such digital assets after your death, for example, personal email accounts, not those within the corporate domain, which are neither inheritable nor owned by the employee. It is based on blockchain technology due to the intrinsic need to govern hereditary information. These needs include a public and secure record of transactions that cannot be altered, confidence in the integrity of stored data, decentralization of information, the immutability of records, efficiency, and speed. These features empower users with greater control over their data and transactions. Users, as beneficiaries of their digital assets, will be able to establish specific, planned, and transparent conditions for their digital assets after death safely and transparently, ensuring that the records of these transactions are secure and auditable. By adopting these solutions and planning their digital estate in advance, users can ensure that it is managed securely and effectively for future generations.

1. Introduction

The inheritability of digital assets is only sometimes clear and may vary depending on the platform and current legislation. In Spain, the heritability of digital assets is regulated by various regulations and laws that address specific aspects related to privacy, data protection, and intellectual property.

Today, a significant critical mass of mature Internet users are beginning to wonder what their assets will become when they pass away [1]. Through traditional mechanisms, they can ensure the fate of the analogue assets they have accumulated. However, similar instruments are not available for digital assets, which often acquire great sentimental and economic value because the contracts signed with digital service providers or the policies they have in place usually do not provide for what happens when a person dies. Therefore, by its very nature, this digital estate should be managed more dynamically and customizably than the analogue estate, which is traditionally tied to more static and complex processes.

A digital legacy can be defined as the corpus of electronic data that users leave behind, either in a physical medium or on the Internet, when they die [2]. It includes profiles on social networks, all types of online accounts, email inboxes, cloud memories, some types of licenses, chat histories, and media. A person's digital wealth is born when he/she enters into a contract with a service provider to provide digital services.

Whether opening a social network account, Internet banking, email, cloud storage, or similar digital assets, it is enriched through the user's interactions with the service. It consists of all goods, rights, and digital assets: photos, videos, movies, e-books, access credentials to digital services (email, banking, storage, among others), digital money, subscriptions (magazines, newspapers, and similar resources), and licenses accumulated by the user throughout the use of digital services.

According to ONTSI's report [3], which studied the use and attitudes of digital content consumption, the term digital content is defined as the entirety of information produced and stored in an electronic format that is commercialized, transmitted, and reproduced through telecommunications networks and Information and Communication Technology (ICT) services. This report is the latest report published by ONTSI, capturing the main trends in digital content consumption habits in 2021 and their evolution over the past five years. This report focused on the population aged between 16 and 74 years and concluded that 76.4% use educational digital content, 82.2% use digital photos and images, and 80.3% watch movies, series, and digital videos. Moreover, it indicated that content consumption has become normalized and widespread among the Spanish population, with almost everyone consuming at least one type of digital content, showing significant growth since the previous study in 2016 [4]. According to the report, this growth was further intensified by the impact of COVID-19, which led to a 39% increase in

* Corresponding author.

E-mail address: rpastor@scc.uned.es (R. Pastor-Vargas).<https://doi.org/10.1016/j.bcr.2025.100289>

Received 1 February 2024; Received in revised form 4 February 2025; Accepted 10 April 2025

Table 1
Digital assets categories.

Category	Digital assets
Email	Email accounts/contact data/messages or conversations contained in email account
Self-produced multimedia	Digital files uploaded to platforms and social networks, like photos, videos, music, e-books, and movies
Multimedia subject to third-party intellectual property	Digital files purchased from stores; contents are similar to the self-produced category
Credentials, coupons, discount codes, and contacts	Contacts on platforms followers (especially important for influencers)
Hardware	PCs, laptops, smartphones, tablets, external hard drives, USB devices, e-book readers, and video game consoles
Financial electronic accounts	Cryptocurrencies: Bitcoins, Ethers, etc. / Prepaid cards (balance for use in: phone calls): online banking, payment services (PayPal, Google Pay), online stores (Amazon), cryptocurrency platforms (Bitcoin), streaming services (Netflix, Spotify), and Google or Apple accounts
Licenses and properties	Software licenses (image editing programs), video games (Steam, Origin), and belongings and avatars in online games
Software as a Service (online platforms)	Cloud services, work platforms (Slack, WordPress), social media tools (Hootsuite), financial software (NetSuite), crowdfunding platforms (Patreon, Kickstarter), and YouTube channels
Social media profiles	Facebook, Twitter, Instagram, LinkedIn, WhatsApp, Google, SnapChat, Skype, YouTube, and Dating sites

consumption frequency during the pandemic, trending towards becoming a structural habit.

The main devices used for consumption are smartphones at 76.7%, desktop or laptop computers at 64.7%, tablets/iPads at 33.3%, and smart televisions or televisions connected to the Internet at 31.1%. Among the 12 types of digital content analysed, the most commonly used are digital photos, applications, digital presses, social networks, movies, series, documentaries, videos, music, podcasts, reading websites, and blogs and forums.

The massive adoption of social networks, digital storage platforms, and digital services in general has contributed significantly to users generating a unique personal digital footprint on the network, voluntarily or otherwise. Users are evolving in their behaviors and tastes. However, the digital footprint remains on the network. The new digital needs of users include the ability to control their footprint. Table 1 presents the leading digital assets a person should consider in their will, broken down by categories.

The management of these digital assets is carried out manually on each source, as each social network has its own privacy policy and offers different services. However, all provide management options after death. For example, Google identifies a deceased person's accounts as inactive until appropriate measures are taken. It gives the right to decide what will happen to the account if it is not used due to death or an accident. Google provides a specific form in its Help Center for reporting the death of an account holder. In the case of Facebook, it allows a profile to be deleted or turned into a memorial account if requested by relatives. It even allows this in case the user has medical incapacity. A person close to the deceased can directly contact the company, but will need a copy of the account owner's death certificate or a legal document proving the requester's right to represent them. PayPal has the only option to cancel the account. The account balance becomes part of the deceased's digital estate and inheritance. Although from a tax perspective, since 2015, cryptocurrencies such as Bitcoin have been considered official currencies and tax-free (not in most jurisdictions), and the inheritance of cryptocurrencies has not been uniformly regulated [5].

In practice, the deceased's cryptocurrency accounts and balances are transferred to the heirs or executors named in his digital will, along with the rest of his digital estate. Legal ownership can be transferred, but balances cannot be transferred without access to the private keys. However, no clear and homogeneous legal regulation exists on how cryptocurrency inheritance should be managed and transferred.

This issue generates some uncertainty and complexity, as cryptocurrencies, being decentralised and without a centralized issuer or intermediary, require specific mechanisms for their transfer and management after the owner's death. Therefore, the need for uniform regulation in this area is one of the main limitations and challenges users face when planning the inheritance of their digital assets in cryptocurrencies.

Instagram manages digital inheritance similarly to Facebook. The help section explains how to report an account belonging to a deceased user. Instagram allows for an account to be memorialized or

permanently deleted. Twitter offers these services, but with different procedures. Direct relatives or others affected can request the removal of certain content related to the deceased, such as images or videos, through a privacy form. Finally, Gmail or Outlook allows the protection of a user's email information and its management or deletion by a second person after the user's death.

Instagram allows for an account to be memorialized or permanently deleted. Twitter offers these services, but with different procedures. Direct relatives or others affected can request the removal of certain content related to the deceased, such as images or videos, through a privacy form. Finally, Gmail or Outlook allows the protection of a user's email information and its management or deletion by a second person after the user's death.

The transfer or inheritance of various digital assets or profiles has legal implications that still need to be fully resolved at present [6]. For some potential issues related to the inheritance of digital assets, such as intellectual property management, blockchain [7] could be a solution element. Blockchain does not intrinsically provide proof of ownership, as it is a technology, not necessarily a legal record.

The initiative discussed in this paper, in conjunction with collaboration within the Alastria ecosystem [8], aims to contribute to the evolution of legislation to facilitate appropriate adaptation.

The new digital models of the commercialization of goods enable the management of emerging needs and various aspects of digital interaction with assets [9]. For example, introducing regulations such as the General Data Protection Regulation (GDPR) [10] empowers users to have a say in managing their data.

Additionally, to the rational arguments presented before, personal or sentimental reasons promote the need for mechanisms to activate the deletion of the digital footprint. For this purpose, many platforms are adapting their services to meet these needs. For instance, Facebook offers mechanisms to convert a user's profile into a memorial in the event of death [11].

Considering the above, there needs to be a methodology or reference framework to define the roles involved in managing digital assets and their participation in such management. It is also necessary to provide secure and reliable mechanisms for the verification and validation of the ownership of such resources. Blockchain can provide these mechanisms, which intrinsically provide these features and track the assets of the digital inheritance and their property afterlife. Then, this paper proposes a blockchain-based framework that provides these features and structures the management of users' digital legacy.

The paper is divided into several sections that provide a holistic vision of the proposed solution. Section 2 presents the expected features in managing digital legacy and actor-involved areas. Section 3 presents the actual works in this area, showing some related papers that offer a global or partial view of the problem. It also examines current national and international regulations regarding digital inheritance. Section 4 describes the solution in terms of its functional and technical aspects. Section 5 breaks down the technical results achieved and the volumes

associated with the various tests conducted. The final section outlines a potential roadmap for improving the solution and implementing additional functionalities.

2. Features and domains in digital legacy

Proper management and preservation of this digital inheritance are becoming crucial for legal and economic purposes [6]. It is emerging as a new digital necessity for the service. Currently, lawyers specializing in new technologies suggest that the will should mention what the heirs should do with the digital estate. They also recommend including passwords and keys that the testator must access to obtain information [12].

The solution must accommodate a will similar to the conventional one. However, with a distinguished difference, it works on digital profiles and the accumulated digital estate or digital footprint generated by individuals throughout their lives, including consents regarding it.

The digital legacy implementation must consider a unique framework/solution not exist in the market, as will be shown in the next section, which will help to fill the current gap in the provision of digital services adapted to offer customizable control over individuals' digital assets. It must provide a robust solution with a reliable and easily integrable system for managing estate, profiles, and digital assets, especially those critical for death or disability. The above characteristics are the main features that must be considered in any solution implementing a digital legacy. In particular, blockchain is the backend manager that implements these features and needs. Blockchain is a technology that provides reliability and integration inside a secure environment for tracking several types of resources [13].

There are some essential domains in which the impact of correct management is particularly relevant, providing clear opportunities for using digital legacy frameworks. Some of these domains are as follows:

- Customer segment. Insurers could offer a new exclusive digital support service within their death, disability, or other insurance policies, allowing users to define the future of their digital assets and profiles. Also, insurers could implement automatic management of disability insurance. When the system is fed with the user's disability information, this framework solution could significantly simplify bureaucratic procedures related to powers of attorney, asset transfers, and administrative tasks.
- Social networks or digital platforms. Integrating this functionality directly into their settings could provide added value to their users. It avoids the manual management of transferring control of their profiles in cases of death or incapacity. Most social networks used in Spain are foreign, so this target market would be part of the internationalization strategy for exploiting our solution as a standard service to integrate [14]. For digital platforms, broadly defined as platforms where users can register and create a digital profile to access potential products and services, the proposed solution would allow for managing the user's profile as an asset and configuring its inheritability parameters. Thus, users could define their will through a blockchain-based smart contract and establish their wishes regarding the management of their profile on the relevant social network or platform. It is important to differentiate between mere configuration and establish a smart contract. The latter is not a legal contract [15] in the same way as a simple configuration parameter, but it benefits from the security of immutability and cybersecurity provided by the blockchain network. Some smart contracts can be considered legal contracts, depending on the jurisdiction and how they are structured. For a smart contract to be regarded as legal, it must meet certain requirements, such as:
 - Consent: The parties must consent to the terms of the contract.
 - Capacity: The parties must have the legal capacity to enter into a contract.
 - Lawful object: The subject matter of the contract must be legal.

- Formalities: In some jurisdictions, certain contracts must comply with specific formalities (for example, being in writing). If a smart contract meets these criteria and if the involved parties accept it, it may be legally recognized. However, the regulation of smart contracts is still evolving in many areas, which may complicate their legal recognition.
- End users must have a reliable, legal, and highly accessible digital method for managing and transferring their digital assets [16]. Users could define their will regarding their assets in the case of death or incapacity and in scenarios like divorce, childbirth, graduation, or retirement.
- Public administrations will be interested in offering this service as part of the functionalities of a citizen's portal or electronic administration.
- Legal service firms, lawyers, or law firms need to incorporate the digitalization of their service offerings and provide a link between traditional legal services and those of the future. In this case, they can act as triggers of the process by certifying a person's death.
- Hospitals or medical services, like the above, can initiate the death certification and thereby trigger a workflow similar to that of COVID-19 digital medical passports and immunity certificates [17].

After analysing the above needs in the different domains, it can be concluded that it is necessary to provide a digital solution that implements functionality similar to a traditional will but with the unique aspect of addressing the profiles and digital assets gathered over a person's lifetime. Considering the previous features, using a technology like blockchain seems logical. So, this blockchain solution must implement the following set of characteristics:

- Leverage smart contracts [18] to ensure the tamper-proof execution of the system users' will regarding the disposition of their digital assets accumulated throughout their lives.
- Underlying this, the solution facilitates the fulfilment of users' wishes regarding digital services such as social networks, storage platforms, and banking services. This feature ensures what will happen to their profiles and digital assets in case of death or temporary incapacity (account closure, digital inheritance).

Next, the existing solutions or frameworks will be analysed to check whether they meet the above needs and develop the main objectives of implementing the digital legacy.

3. Related work

In the previous section, we presented the necessity of a blockchain-based solution for managing an individual's digital assets and estate throughout their life, ensuring privacy, security, customizability, and easy integration, and facilitating the management of their digital footprint/inheritance and subsequent actions after their death. Based on this, this section's state-of-the-art study shows the current digital inheritance (or digital content) state.

The study is carried out separately in the academic and regulatory fields. In this way, it is possible to have a global perspective of the regulatory implications associated with the proposed solutions found in the literature.

3.1. Blockchain domain review

There are currently some ideas in the market attempting to address this problem by including access keys to websites [19] in traditional wills, which needs to more holistically solve the presented problem. We do not observe studies that openly address this issue. Several articles provide partial insights into potential architectures, storage, security,

and integration, but not into digital asset management, governance, customization, delegation, and legacy.

We observe studies focusing on defining versatile architectures around blockchains and implementing some previous features. However, there are no blockchain applications in this particular area of digital legacy. For example, a blockchain architecture for industrial applications [20] focuses on industrial applications of blockchains. It compares public and private blockchains while proposing a comprehensive solution based on Ethereum to implement a decentralised application. However, it does not address customization, integration, or digital assets. Regarding Blockchain in the financial sector (FinTech), Baliker et al. [21] presented the use of blockchain technology in FinTech and highlighted open research challenges and future research directions in blockchain-based FinTech applications, particularly in security. Again, it does not address digital asset inheritance, integration, or customization. Cao et al. [22] introduced a comprehensive approach to study the theoretical model that underlay the fundamentals of blockchain. Their methodology encompasses the design of network services, mechanisms, and various algorithms, providing valuable insights into the effective implementation of blockchain technology. Although it introduces the scope of use in the Internet of Things (IoT), it does not define digital assets. The customization and management, which are features to implement in this particular case proposed in the paper, need to be addressed.

The IEEE [23] described the reference architecture of the blockchain-based application in the electronic invoicing business. This provides a foundation for blockchain-based implementation solutions for the electronic billing business, which helps optimize the user experience and business efficiency. This paper provides a reasonable basis for associated architectural recommendations, but needs more content regarding digital assets and solution integration.

Some studies emphasized storage and security rather than digital assets. For example, Feng et al. [24] proposed a Digital Asset Platform (DAP) based on blockchain. It relies on digital signatures and smart contracts to enhance the reliability of supply chain financing between companies and banks, and storage on the blockchain to ensure traceability. It focuses on the banking sector and does not address digital assets or their management in the event of the owner's death. Liu et al. [25] presented an overview of the security of storage and the efficient use of industrial production data, incorporating blockchain to safeguard the privacy of smart factories and data automation processing. Additionally, they designed an algorithm for verification and anonymous storage of production status, known as the Double Color Ball (DCB) algorithm. It focuses on the Industrial IoT (IIoT) and does not cover digital assets, inheritance, customization, or integration with other solutions. Saraiva et al. [26] detailed a web application named MIRIAM based on the blockchain using the Hyperledger platform, which allows decentralised and reliable storage of relevant information necessary for managing medical professionals' records. It also explores the Brazilian regulatory framework to clarify the feasibility of the solution in a real-world workflow scenario. Like the previous case, albeit in another field of use, it does not include the study of digital assets, inheritance, customization, or integration with other solutions.

Some other studies have focused only on security features. Kleinaki et al. [27] showed a notarization service that uses smart digital contracts to secure queries to a biomedical database and their respective results. The aim is to ensure that retrieved data cannot be modified after retrieval and that the database cannot validly deny providing specific data due to a particular query. The study does not encompass digital assets or inheritance within its scope. Yu [28] proposed a blockchain-based Supply Chain Finance (SCF) system that combines smart contracts and the main methods of supply chain financing to address the drawbacks of SCF, using the Ethereum blockchain. Like the previous paper, it provides a good overview of security but does not reference digital assets, their governance, or management.

There are academic references focused on interoperability and reference architectures. Still, none of them, although important for interoperability, address digital assets globally, let alone the management of the underlying inheritance. Scheid et al. [29] defined an interoperability solution with a notary scheme-based blockchain interoperability Application Programming Interface (API), Bifröst, which allows transparent interaction by users or legacy applications with several blockchains. Soares et al. [30] presented a blockchain-based customizable document registration service. The registration service allows the creation of different types of generalized documents for various application domains and stores them in one or more integrated blockchains through APIs. Wöhrer and Zdun [31] presented how designing blockchain-based applications is challenging and requires several coordinated architecture decisions. It provides architectural design decisions and decision options regarding patterns and practices. This research shows that most design decisions are influenced by the need to offset current blockchain drawbacks, such as scalability, privacy, and usability, by using centralized elements. Also, it suggests that hybrid architecture is beneficial in many design situations, such as the one proposed in this paper. Dwivedi et al. [32] focused on smart contracts as a key component of today's blockchains, which are critical in controlling Decentralised Autonomous Organizations (DAOs).

However, smart contracts still need to be legally binding and enforceable; this makes it difficult for businesses to adopt the DAO paradigm. Therefore, this study reviews existing Smart Contract Languages (SCLs) and identifies properties critical to any future SCL for drafting legally binding contracts. Spychiger et al. [33] used the action design research approach to create an incentive system featuring a rating mechanism influenced by data correction measures. This contributes to an understanding of incentive design in inter-organizational blockchain-based data platforms. Furthermore, the authors explored incentive design in permissioned blockchains and discussed their latest implications.

In general, this finding indicates that no solution or proposal in the literature have focused on a digital legacy and implemented the needed features. The only found reference [34] provides an initial vague overview of the problem we aim to address, but it needs to address the management of inheritance and digital asset rights. It defines a blockchain-enabled digital rights management system incorporating a network architecture for sharing and managing online education multimedia resources based on the combination of public and private blockchains.

In summary, comprehensive studies have focused on partial approaches to storage, interoperability, security, and asset management. However, no systematic references are considered initial studies to address the specific problem presented in this paper. The reviewed literature reveals a lack of digital file management systems, an inability to manage social networks, and mechanisms for transferring digital content.

3.2. Regulatory domain review

It should be noted that there is no recognized international standard for digital legacy management. Due to a lack of definition, a specific study was conducted at the national level (Spain), but international regulations from other countries were included. The two levels are exposed in the following subsections.

3.2.1. Spanish regulations

Catalonia, Law 10/2017 of June 27 on digital wills, was the first to address this issue in Spain. However, it provided different solutions compared to the subsequent LOPD [35], not only regarding the persons authorized to act against information society service providers to execute the instructions of the deceased but also regarding the form of the instructions themselves or the actions that can be carried out concerning the digital contents of the deceased. Thus, within the national environment, reference is made to the Draft Law on Digital Wills and

the amendment of the second and fourth books of the Civil Code of Catalonia [36].

LOPD, Law 3/2018 of December 5 on personal data protection and guarantee of digital rights, incorporates the regulation of digital wills into the Spanish legal system. Digital wills are understood as documents of final wishes that allow us to leave instructions on the fate of our digital content in case of death.

This law achieves national regulation of the rights of legatees and respect for the wishes of deceased persons concerning their digital legacy.

3.2.2. International standards

In the case of Europe, digital legacy regulations are closely related to the European GDPR (RGPD) [37]. Regulation (UE) 2016/679 of the European Parliament on the protection of natural persons concerns about the processing of personal data and the free movement of such data, repealing Directive 95/46/EC (RGPD). For example, France [38], according to its new law, in compliance with the GDPR, establishes and regulates the protection of data belonging to deceased individuals. The law grants individuals the right to make preventive decisions regarding the processing of their personal data after death.

There is also existing legislation in various states in the United States of America (USA). To list a few examples:

- Connecticut [39] establishes and regulates access to the email accounts of deceased individuals.
- Delaware [40] regulates access to digital assets and accounts by fiduciaries.
- New York [41] provides access to a decedent's electronic mail, social networking, and/or micro-blogging accounts to the executor or administrator of the decedent's estate.

In conclusion, Europe and the USA are concerned with digital legacy and its legal regulation. These regulations make a reliable solution for managing digital assets generated or owned by deceased persons even more necessary.

4. Solution description

In the previous section, it shows that there are no solutions available for the needs raised for the management of digital legacy, and that specific regulations in Europe and the USA should have a reference framework and implementation. This work presents a framework based on blockchain developed in the context of the TrustForWills project. The solution is a project co-funded by the Ministry of Energy, Tourism, and Digital Agenda (MINETAD) [42] and the European Regional Development Fund (FEDER-ERDF) [43] to develop a solution based on blockchain technology.

Why blockchain? It relies on blockchain technology as a distributed ledger technology that allows for the creation of immutable and transparent transaction records. By using smart contracts on the blockchain, individuals can designate beneficiaries of their digital assets and establish specific conditions for their transfer after death. These contracts provide a secure and transparent way to transfer digital assets and ensure that the records of these transactions are secure and auditable.

The solution thus offers an innovative approach that addresses the issues associated with digital asset management and proposes an efficient way to transfer digital assets after a person's death. By implementing these solutions and proactively planning their digital estate, the aim is to ensure secure and effective management for future generations.

This project identifies the need to regulate the legacy formed by digital files derived from an individual's activity in digital environments after a person's death. The developed system provides a reliable and easily integrable solution by being based on technological standards, such as RESTful, for managing estates, profiles, and digital assets, as it is based on emerging technologies. Therefore, the platform supports three

clearly differentiated use cases or functionalities based on blockchain technology.

The management features are divided into three categories: digital profile management, inheritance management, and consent management.

The inheritability of these assets is only sometimes clear and may vary according to the platform and current legislation. In Spain, the inheritability of digital assets is regulated by various regulations and laws that address specific aspects related to privacy, data protection, and intellectual property. Among them, the following stand out:

- GDPR [10]: GDPR is a European Union regulation that establishes rules for the protection of personal data. Although it does not directly address the inheritability of digital assets, it provides guidelines on the processing of personal data after the death of an individual.
- Organic Law 3/2018 of December 5 on Personal Data Protection and Guarantee of Digital Rights (LOPDGDD): It supplements the GDPR in Spain and specifically addresses issues related to personal data protection and digital rights. It contains relevant provisions regarding digital data's inheritability and privacy management after death.
- Intellectual property law [44]: Spain's intellectual property law regulates copyright and other rights related to intellectual property. If digital assets include copyrighted content, this law may be relevant to determine their inheritability and posthumous use.
- Legislation on wills and successions [45]: Laws governing wills and successions in Spain may also affect the inheritability of digital assets. Such legislation provides relevant information on how digital assets are managed in the context of succession and asset distribution.
- Law 30-1991 of December 20 amending the civil code on wills [46].
- European Commission. Successions and wills. EU rules on international succession [47].
- Digital rights [48]: These rights have been recognized in recent years to improve the regulatory framework adaptation to the digital world, protection, and relationship with similar processes in Europe and the world.
- Banking and financial regulation [45]: In the case of digital assets such as cryptocurrencies or online investment funds, banking and financial regulation may also be relevant. Consulting applicable rules in this area can provide information on how these assets are managed in the event of death.

A combination of platform policies, local legislation, and digital security practices determines the inheritability of digital assets. Understanding these considerations is crucial to ensure a smooth and secure transition of digital assets in the case of death. These regulations are constantly evolving and depend on the state and even communities, so constant study and the solution provided in this paper are necessary.

Digital profile management allows for the selection of actions to be performed automatically in each system integrated into the platform. The following systems are integrated:

- Mail accounts (Hotmail/Gmail).
- Social networks (Facebook, Twitter, LinkedIn, Google+, and GitHub).
- Wallets (Bitcoin [49] and Ethereum. [50])

Particular actions are required for this type of management. These actions involve deleting accounts, users, messages, comments, photos, contacts, etc., as determined by each social network or platform.

At the operational level, the following steps are required:

- A digital executor should be appointed to manage digital profiles and accounts.
- The designation can be made anytime during the asset holder's life.

- The right to erasure is inherently linked with the GDPR. All profiles are closed after death.

Two types of users are distinguished:

- End user: a user who wishes to express their will regarding digital assets.
- Executor user: a user who inherits digital assets.

Conceptually, a distinction is made between the following:

- Digital profiles associated with social media accounts (digital profiles).
- Files hosted in the cloud (inheritance management).

Managing digital profiles presents a more significant maintenance challenge, as providers like Facebook and Twitter continuously update interaction APIs. The use case of inheriting assets hosted in the cloud allowed maintaining the project's core functionality (digital inheritance) without needing continuous changes or updates to the backend services integrated with provider APIs.

Inheritance management. A subscription plan enables one or multiple executors to store categorized multimedia content for future management. It is the equivalent of a storage system but is based on distributed technologies using the InterPlanetary File System (IPFS) [51] protocol and has the ability to delegate the categorized content.

Consent management. A consent management mechanism is implemented in compliance with the GDPR, enabling users to select the consent they grant to each service-providing company (categorized by sector). Clearly, this mechanism extends beyond the functionalities of the Robinson list [52], a blacklist of individuals or companies considered to pose a risk to the national security or economic interests of a country, maintained by the government of the United States, to ensure consumer rights and to identify, in real time, the contact methods accepted by the user, if any, are approved.

For instance, providers like telecommunication providers (Movistar, Vodafone, Orange, etc.), electricity and gas suppliers, and water suppliers will query the user's permissions (updated in real-time) via Representational State Transfer (REST) API [53] before contacting them (if no consent is granted, they may contact the user by phone in the mornings or use other communication mechanisms).

This API accessibility would facilitate providers' implementation of user-defined pre-authorizations through smart contracts, thus ensuring a secure and immutable authorization process and covering a market scope not currently contemplated.

4.1. Technical architecture

The solution framework is built upon Ethereum, an open-source platform using blockchain technology [54]. This selection is primarily due to standardization efforts facilitated by the Enterprise Ethereum Alliance (EEA) [55]. The EEA is a consortium comprising over 100 companies from diverse sectors, working collectively to enhance the privacy, security, and scalability of the Ethereum blockchain. These efforts aim to make it more suitable for application in corporate environments. Noteworthy members of this alliance include prominent companies like BBVA, Banco Santander, Microsoft, Accenture, Deloitte, ING, and Credit Suisse.

It uses cloud computing technologies to ensure the global availability of the service and decentralized storage technologies (IPFS). With them, we ensure the availability of critical digital data or assets, such as personal photos, for inheritance.

An essential aspect is the storage and registration of keys and where. Data tuples are stored to model the correspondence between digital assets, usage permissions, and executors (emails). By default, the executor is not registered in the solution, but the system includes your email as

another option in the will. Once the inheritance process is triggered, the executor receives an email to register and create an account in the solution. In this way, the executor can register as a regular end user. Passwords themselves are not stored anywhere; the solution is designed as an authentication token manager. The system proposes several solutions in this regard: Single Sign-On (SSO), token-based authentication (by default), and Public Key Infrastructure (PKI); in the future, it will integrate AI solutions for biometric authentication or federated identity management. One significant challenge is maintaining the authentication token alive indefinitely. Periodic token refresh mechanisms offered by Facebook and Twitter are employed for this purpose. A Role-Based Access Control (RBAC) mechanism determines a user's role within the system.

The end user is responsible for sealing this information by storing it in the blockchain using their account and invoking their smart contract. Smart contracts are programmed to act upon the occurrence of triggering events, such as death or temporary incapacity—simulated with an application admin user button.

When the action-triggering button is pressed, the smart contract is activated, allowing execution of the permission issued by the end user (close account, renew access token, etc.). Only the end users can edit their will by invoking their smart contract.

The external world (action-triggering button) signifies only the evolution of situations to be evaluated (death or temporary incapacity); smart contracts interact within the blockchain to ultimately execute web services, which in turn interact with digital assets.

In this way, the receiver acts as an executor and will only have access after death. This action has the advantage that indiscretion or loss of user information does not jeopardize the inheritance process. A token is used to avoid the security problem of stored keys. The application has been designed so that keys are not necessary to store; only the executor's email is stored. Before the final user passes away, they include the executor's email address. The executor receives a link to execute the inheritance when the user passes away. The final user is the only one with a key and password, with double authorization via tokens.

The implementation has been executed on Alastria (Quorum) with a replica on a local Ethereum network, which uses the Proof of Authority (PoA) consensus algorithm to prevent potential information losses associated with migrations. This alternative has proven effective, considering that in the initial phase of the project, the Alastria network (formerly Lyra network) is in its Arrakis [56] version. In contrast, it operates on the current T-network (without backwards compatibility). In Fig. 1, the technical architecture components are presented.

This replica's primary purpose is to prevent data loss that could occur during migrations between different versions of the Alastria network. The Alastria T-network leverages Quorum technology [57], an enterprise version of Ethereum, making it compatible with Decentralised Applications (DApps) developed for Ethereum but with additional privacy mechanisms provided by Quorum. Quorum employs advanced mechanisms to ensure the privacy of transactions and contracts, making it a reliable choice for enterprise blockchain applications requiring high confidentiality. In many systems, nodes may have different views of the data due to replication, partitioning, or the nature of distributed storage. This difference in nodes means that even when a quorum is present, not all nodes necessarily have the same information. This network consists of around 120 nodes from different entities, rendering it stable since January 2019, enabling the deployment and testing of smart contracts. Therefore, the Alastria T-network is a public-permissioned blockchain platform aiming to establish a legally effective service ecosystem in Spain, using Quorum technology and counting on the participation of over 500 partners from various sectors.

The dynamic generation of smart contracts for managing digital profiles, digital inheritance, and consent is carried out using recognized templates and patterns. It stores the smart contracts defined in higher layers to be invoked when needed. The advantage is that by storing the

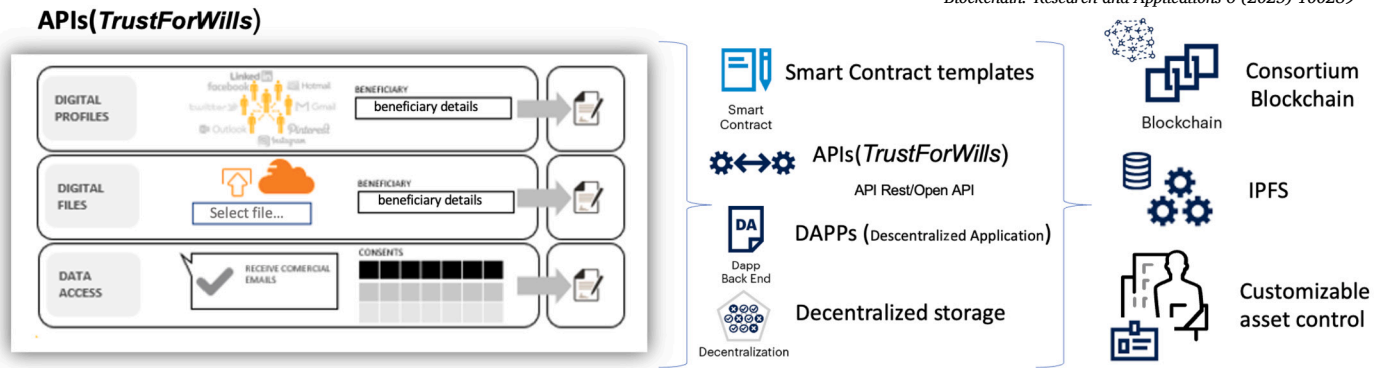


Fig. 1. Digital legacy blockchain platform architecture and components.

contract on a blockchain, all parties involved in managing digital assets can ensure that it has not been manipulated.

We implement the contract factory concept [58]. A smart contract factory is a design pattern used in blockchain technology to efficiently and automatically create and deploy smart contracts. It is a template or model for creating smart contracts using configurable parameters. Regardless of their programming experience, it enables anyone to easily and quickly create and deploy smart contracts.

The solution, which is based on a private/consortium Ethereum network with distributed storage capabilities, is expected to address the following needs:

- Private/Consortium: Record writing will be exclusive to consortium members, allowing third parties to query, excluding general public interaction with the network.
- Immutability and data integrity: Ensuring that stored records are cryptographically secure without compromising their integrity.
- Scalability: Guaranteeing that increasing network transactions does not compromise network synchronization, security, accessibility, or data integrity.
- Security: Providing support for data encryption and access restriction capabilities for participants and third parties.

These capabilities ensure the solution's scalability, compliance with established legal regulations, and privacy for citizens.

In an environment like the Internet, where new digital services emerge daily and are capable of becoming part of citizens' digital heritage, we have given special attention to the platform's integration capabilities. This is why the solution's framework is interoperable and integrable via REST APIs based on *OpenAPI v2.0* [59], enabling interaction with other systems. Consequently, the platform is interoperable for third-party usage in Software as a Service (SaaS) mode. This interoperability stands as one of the fundamental pillars of the platform.

The processes for managing digital accounts on services like Facebook and Twitter and email providers like Gmail and Outlook have been automated. Digital banking services and insurance management could be considered in a subsequent phase.

The digital heritage management platform also possesses the capability to access information available from other services. To achieve this objective, appropriately secured communication interfaces have been developed, enabling third-party usage of the platform. We have studied the advantages of implementing these interfaces using Google Remote Procedure Call (gRPC) over HTTP/2, allowing us to benefit from features such as the utilization of a single connection, compression of headers, and server push services.

Additionally, the use of OAuth2 [60] as a third-party authorization method is mandated.

Given that it is not advisable to store sensitive information on the blockchain layer and that storage capacities within blockchain blocks are limited in Ethereum, information must be stored in a distributed manner that different members can access.

Most of the data are stored in a private data store functioning as a distributed relational database. The data store is configured to perform automatic signatures of transaction sets onto the public chain in bulk at any required interval.

The application layer contains all the application's business logic and connections with third parties through technologies like REST, Simple Object Access Protocol (SOAP), or gRPC.

This layer uses the information stored in the data storage layer and the blockchain layer and initiates the execution of the smart contracts stored in the blockchain layer.

The user interface layer is decoupled from the backend and communicates via the REST API.

The solution is physically deployed within a cloud provider (AWS) that complies with current data protection legislation.

Regarding development, the solution is shown in Fig. 2. The main components of the solution are structured as follows:

- A frontend in Angular, separated from the backend, communicates via the REST API.
- A Java backend is developed under the microservice paradigm using the SpringBoot framework, with a relational database, MySQL, following the Master-Slave model.
- Interaction with the blockchain occurs through a REST API exposed in an API gateway, implemented using Lambdas in Go to establish direct communication with various blockchain networks (public Ethereum, Alastria, and private Ethereum) and the IPFS network.
- Storage is carried out using a non-relational database (DynamoDB).

The solution is designed in three layers (Fig. 3):

- Blockchain layer: This layer is based on Ethereum. It stores the smart contracts defined in higher layers to be invoked when needed. It offers the advantage that, with the contract stored on a blockchain, all the actors involved in managing digital assets can ensure that it has not been tampered with.
- Data storage layer: This layer is used for data storage, as the storage capabilities within blockchain blocks are limited in Ethereum. The solution needs to be able to store information in a distributed manner that may need to be queried by different consortium members. Most data are recorded in a private data store that acts as a distributed relational database. The data store is configured to sign transactions automatically set onto the public chain in bulk at any required interval.

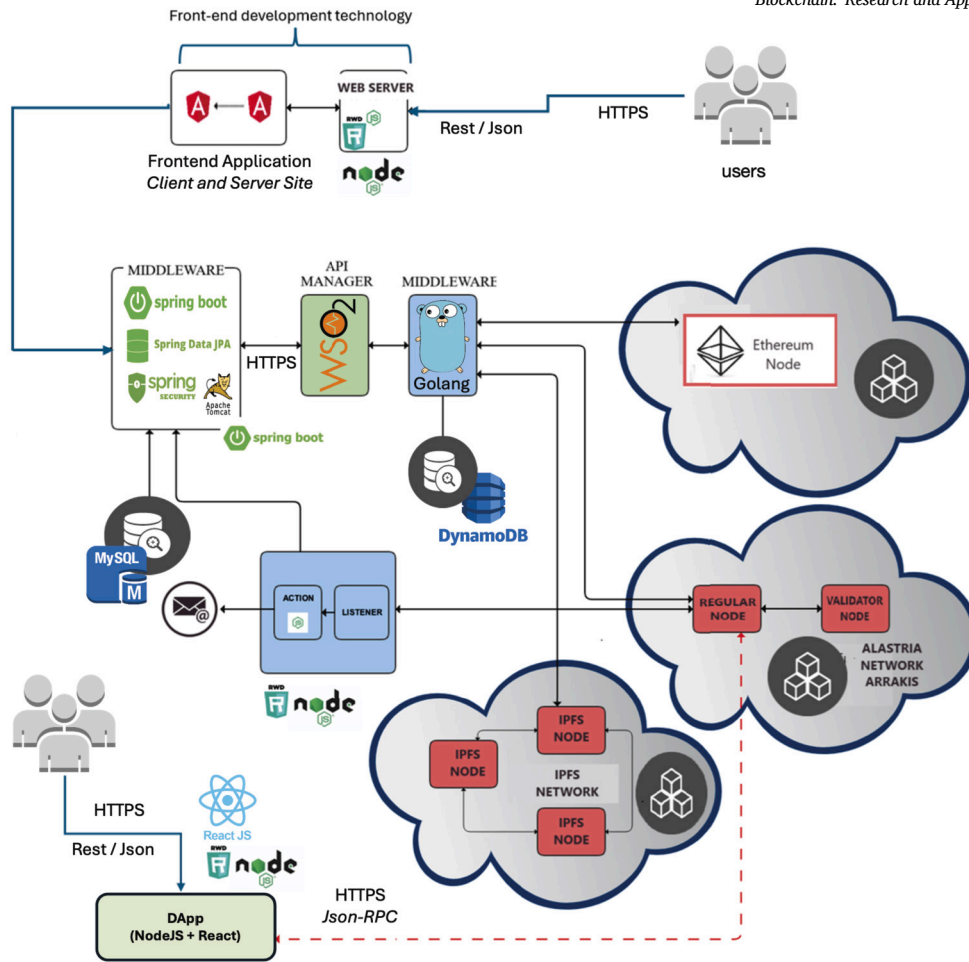


Fig. 2. Developed blockchain platform's components.

- Application layer: This layer houses all the application's business logic, user interfaces, and connectors with third parties using technologies such as REST, SOAP, or RPC. It uses the information stored in the data storage layer and the blockchain layer and initiates the execution of smart contracts stored in the blockchain layer.

4.2. Computational environment of the platform

To achieve this, Alastria has been employed as a blockchain network, a consortium network, and Ethereum-compatible due to its low latency and block generation time of less than 1.5 s. It has a number of transactions per block and a performance exceeding 1500 transactions/s, with a backup on a private Ethereum network using the POA algorithm.

Additionally, the IPFS has been used as a distributed storage technology because it is the most mature and stable in a production environment (at the time of analysis).

Golang has been used as the programming language to develop back-end and blockchain interactions.

Likewise, IPFS has been employed as a distributed storage technology because it is the most mature and stable technology in a production environment (at the time of analysis).

A framework of static analysis tools for smart contracts was created and executed for underlying testing on Docker.

We relied on Solidity's [61] testing framework, providing a structured and automated environment to test smart contracts written in Solidity, the programming language used in Ethereum and other blockchain platforms. This enables test automation to facilitate repet-

itive test scenario execution, support assertions and expectations to verify expected contract behavior, integration with development tools for greater efficiency in the development and testing process, and simulation for different network conditions and execution environments.

The Docker Community Edition (CE) engine [62] was used to execute Docker images, creating containers. Subsequently, the smart contracts were scanned for vulnerabilities, and the results were dumped into the Docker terminal.

The following technologies were used: NodeJS, Truffle, Ganache, Lite-server, Web3, JQuery, and Metamask.

Each smart contract was individually tested and loaded into the Docker image. Smart contracts were not altered from the original contracts.

The Solidity version used in Docker images was the same as the smart contracts, version 0.4.24. Docker images were extracted from Docker Hub and executed in Docker as containers.

4.3. Methodology followed in the development solution

Solution development followed an agile methodology. Corresponding acceptance tests were established for each user story, and as the solution development progressed, a pool of regression tests was defined. The goal of acceptance tests is to verify the acceptance criteria of each user story.

The quality threshold for code during development to approve software delivery is as follows: 0 bugs, coverage above 80, duplicate lines less than 5, and 0 vulnerabilities.

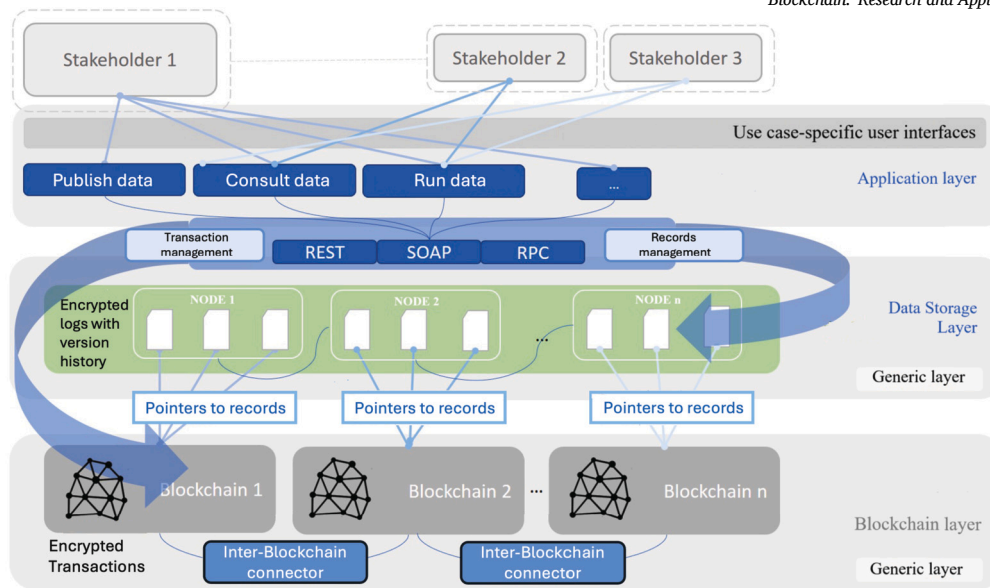


Fig. 3. The solution in three layers.

The static analysis testing scenario was not designed to detect all possible vulnerabilities; the objective was to find potential data result inconsistencies.

The solution does not present critical and high-level software defects. In subsequent releases, medium and low defects are left for resolution, which do not affect the required functionality and are mostly related to interface or minor usability issues.

5. Results and evaluation

All the initially set objectives have been achieved, which essentially involve applying blockchain technology to manage the testament associated with digital assets and the digital profiles of an individual:

- The solution accommodates a testament that operates on digital profiles and the accumulated digital estate, or digital footprint, generated by individuals throughout their lives, including related consents.
- The work represents a unique framework/solution that is currently nonexistent in the market, contributing to filling the existing gap in providing digital services tailored to offer customisable control over individuals' digital assets.
- Therefore, it is a solution with a reliable and easily integrable system based on technological standards such as RESTful to manage estates, profiles, and digital assets—a management aspect particularly critical in the event of decease or disability.
- Upon achieving the primary project objectives from both the business and user perspectives, it provides suitable management for preserving digital inheritance, covering the legal and economic aspects of any digital estate inheritance.
- The solution facilitates the fulfilment of users' preferences concerning digital services such as social networks, storage platforms, and banking services, ensuring the management of their profiles and digital assets in the event of decease or temporary incapacity (account closure and digital inheritance).
- Through smart contracts, the system ensures the tamper-proof execution of users' desires concerning allocating their digital assets accrued throughout their lifetime.
- With blockchain, the system, through smart contracts, automates executions based on events within the blockchain.
- It has been developed as an easily integrable platform/solution for potential clients, providing security and confidentiality in its execution.

This platform expands the range of its services and tailors them to digital businesses.

The primary objective of resolving the current issue arising from the absence of a reliable and easily integrable system for managing heritage, profiles, and digital assets, particularly critical in the event of death or disability, has been achieved. Therefore, TrustForWills effectively acts as a conventional will for individuals' digital profiles and accumulated digital assets throughout their lifetime.

In a production environment, the highest volume of transactions occurs during user onboarding and initial configuration when the user inputs information into the system. Once this information is loaded, only write operations to the IPFS module (content uploads) are expected. Thus, the majority of transactions will be for reading purposes.

Regarding functional and user acceptance testing, verification/certification testing has been conducted using the solution by approximately 750 users. Exploratory analysis akin to crowdtesting. A pool of functional tests described in Jira (Atlassian) in Zephyr is available, previously guided by testing/QA teams. Testers use them to explore the software flexibly, testing various features and scenarios to identify potential issues, encompassing all functionalities denoted in the solution, and involving the use of all phases developed for digital asset management. A System Usability Scale (SUS) questionnaire consisting of ten questions regarding usability and solution value is defined. The satisfaction rate is 93.25%.

5.1. Performance tests

The smart contracts were initially designed to support between 2,000 and 3,000 concurrently connected users with an average of five transactions per day, resulting in an estimated 15,000 transactions per day. Most transactions occur when users input information into the system, as only write operations are expected for the IPFS module (content upload) after loading the data. Most transactions would be reads. The three smart contracts tested with this scenario resulted in the expected returns. These tests are aimed at measuring the end-to-end response time of the solution, not specific component testing.

LoadRunner (MicroFocus) [63] is used to simulate thousands of concurrent users accessing the application through test scripts/scenarios via the solution's front end. The test configuration is set above the previously stated needs with the following load configuration:

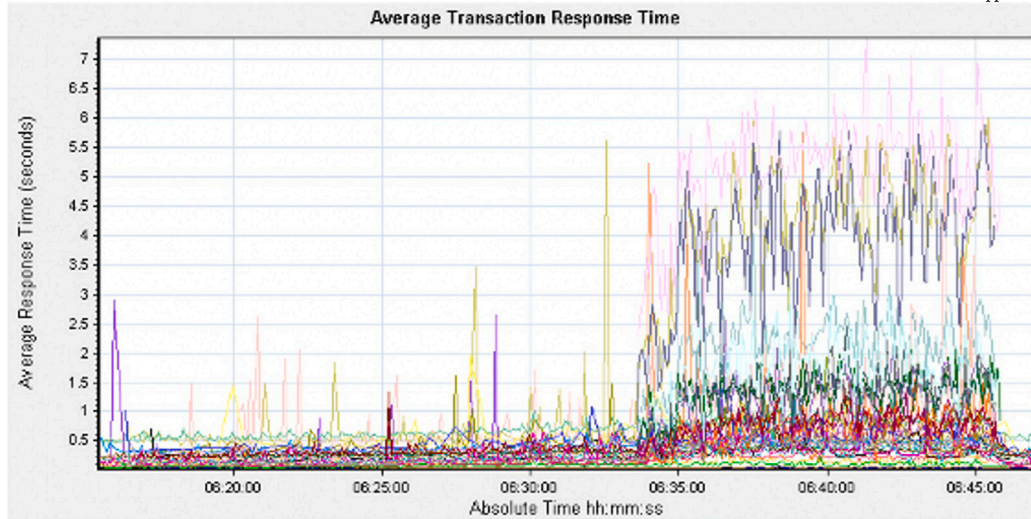


Fig. 4. Response time statistics.

Table 2
HTTP response codes.

HTTP response code	Total responses	Response rate per second for each code
200	8,038,177.0	4223.9
201	5696.0	2993.0
301	44,124.0	23,187.0
302	36,191.0	19,081.0

- Duration: 31 min and 42 s.
- Ramp-up of 2,000 users in 13 min. The process gradually increases the number of users to 2,000 within 13 min.

The statistics related to the response time are shown in Fig. 4, as users are injected into the system.

Each line on the graph corresponds to the functionality stressed with LoadRunner through a script. The x-axis shows the evolution over time, while the y-axis represents the response time. The system provides an optimal response time for the number of users introduced into the system. It is observed that these times worsen when we increase the number of users for the stress tests, tripling the number of users in the system. The test is conducted with LoadRunner installed on-premise, showing a high consistency of the solution. Through root cause analysis of some errors, it is estimated that they are due to infrastructure-related HTTP output issues (Table 2).

How many end users can the application have, and how many concurrent users? The blockchain network can support a proven concurrency level of 2000 users.

Thus, we can extrapolate the following:

- With 3000 users averaging five transactions/day, we obtain approximately 15,000 transactions/day.
- With 20,000–30,000 users averaging five transactions/day, we obtain $30,000 \times 5 = 150,000$ transactions/day, which are still perfectly manageable by the platform.

In a production environment, the highest volume of transactions occurs during the user's onboarding and initial configuration. Only write operations are expected for the IPFS module (content upload) once the information is loaded. Therefore, most transactions are of type read.

6. Conclusions and future work

Unfortunately, the management of digital assets in the event of death and the rights of the deceased person and the legatees are not considered

beyond the existing regulations. No solutions or processes available can implement the necessary characteristics for this type of management. This paper proposes a reference architecture and framework that enables digital asset management to address the need for solutions in this area. The solution incorporates the necessary actors and domains to implement this management properly. This paper shows the architecture and the different management modalities, including compliance management. This management is a relevant step, equivalent to a similar step that must be accessed for the resources of an inheritance in a physical will. From a technological point of view, blockchain technology and distributed storage ensure the solution's reliability, transparency and security, and the storage associated with the digital assets of the digital legacy. This technical architecture is shown in detail, presenting volumetric data to ensure optimal performance from a technical point of view.

In future work, several potential improvements for the solution can be made. The first corresponds with expanding the range of services integrated into the platform, including integration with other players, such as managing digital banking services and insurance. The processes related to the blockchain platform are automatic, but it can be noted that involving lawyers and notaries to oversee the will and proof of life process is essential. When a user registers in the system, their account ownership is authenticated through a verification email and verification from specific platforms like Google Workspace or Microsoft. From a technical point of view, achieving full interoperability with other blockchain networks is critical. It also includes a user's Sovereign Digital Identity (SSI) [64], which has legal validity. We are studying the last two points using SSI based on Decentralized Identifiers (DIDs) and technologies such as Polkadot and Substrate. Again, from a purely technical point of view, the project is implemented using OpenAPI v2.0; a potential improvement involves its migration and technological adaptation to OAS 3.0.

Similarly, consensus among nodes is critical for validating transactions and maintaining network integrity in decentralised networks like blockchain. Implementing efficient consensus algorithms, such as Proof of Stake (PoS) or PoA, can enhance scalability by reducing the resources required to validate transactions. A constant study of legal evolution regarding digital assets and adaptation to the solution is fundamental. Due to their nature as property, the inheritance of digital assets varies widely. Interconnected networks, connecting multiple decentralised networks, can increase the processing and storage capacity available for the network as a whole. This aim can be achieved by implementing bridges or interoperability protocols that allow communication between blockchains or decentralised networks.

CRedit authorship contribution statement

Jesus Hernando-Corrochano: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Rafael Pastor-Vargas:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Roberto Hernández-Berlinches:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Funding

This work was supported by the European Regional Development Fund (ERDF) Program managed by the Ministry of Energy, Tourism and Digital Agenda, within the National Public-Private Cooperation Programme. Strategic Action Digital Economy and Society (AEESD). Call 2017 - Technological Impulse. Project “A Smart contract platform for digital asset management using Blockchain” [grant number TSI-100200-2017-10].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Authors would like to acknowledge the support of the Spanish Government (through the Cybersecurity Institute of Spain, INCIBE), with the Strategic Research Project “Analysis of mobile applications from the perspective of data protection: Cyber-protection and Cyber-risks of citizen information” and the International Chair “Smart Rural IoT and Secured Environments”, within the context of the Recovery, Transformation and Resilience Plan financed by the European Union (NextGenerationEU/PRTR). This contribution is also supported within the framework of the CiberCSI UNED research group and the CiberGID innovation group.

References

- [1] B. Mujtaba, F.J. Cavico, E-commerce and social media policies in the digital age: legal analysis and recommendations for management, *J. Entrepr. Bus. Venturing* 3 (1) (2023) 119–146, <https://doi.org/10.56536/jebv.v3i1.37>.
- [2] D. Reid, Factors that may prevent meaningful digital legacy, in: *Proceedings of the 13th ACM Web Science Conference 2021*, ACM, 2021, pp. 141–142, <https://doi.org/10.1145/3462741.3466670>.
- [3] National Observatory of Technology and Society, Study on the uses and attitudes towards digital content consumption in Spain 2021, <https://www.ontsi.es/es/publicaciones/Estudio-sobre-usos-y-actitudes-de-consumo-de-contenidos-digitales-en-Espana-2021>, 2021. (Accessed 3 September 2024).
- [4] National Observatory of Technology and Society, Estudio sobre usos y actitudes de consumo de contenidos digitales en España 2016, <https://www.ontsi.es/es/publicaciones/Informe-anual-del-Sector-de-Contenidos-Digitales-en-Espana>, 2016. (Accessed 3 September 2024).
- [5] R. White, Y. Marinakis, N. Islam, et al., Is bitcoin a currency, a technology-based product, or something else?, *Technol. Forecast. Soc. Change* 151 (2020) 119877, <https://doi.org/10.1016/j.techfore.2019.119877>.
- [6] J.S. Kharitonova, Digital assets and digital inheritance, *Law & Digit. Technol.* 1 (1) (2021) 19–26, <https://doi.org/10.18254/s123456780015732-6>.
- [7] S. Bonnet, F. Teuteberg, Impact of blockchain and distributed ledger technology for the management of the intellectual property life cycle: a multiple case study analysis, *Comput. Ind.* 144 (2023) 103789, <https://doi.org/10.1016/j.compind.2022.103789>.
- [8] Alastria, Qué es Alastria, <https://alastria.io/>, 2023. (Accessed 3 September 2024).
- [9] A.R. Harish, X. Liu, M. Li, et al., Blockchain-enabled digital assets tokenization for cyber-physical traceability in e-commerce logistics financing, *Comput. Ind.* 150 (2023) 103956, <https://doi.org/10.1016/j.compind.2023.103956>.
- [10] European Commission, Data protection: rules for the protection of personal data inside and outside the EU, https://commission.europa.eu/law/law-topic/data-protection_es, 2021. (Accessed 3 September 2024).
- [11] K.Z. Gach, J. Brubaker, Getting your Facebook affairs in order: user expectations in post-mortem profile management, *Proc. ACM Hum.-Comput. Interact.* 5 (2021) 1–29, <https://doi.org/10.1145/3449248>.
- [12] I.A. Seres, O. Shlomovits, P. Tiwari, Cryptowills: how to bequeath cryptoassets, in: *Proceedings of the 2020 IEEE European Symposium on Security and Privacy Workshops (EuroS&P'W)*, IEEE, 2020, pp. 417–426, <https://doi.org/10.1109/EuroSPW51379.2020.00062>.
- [13] D.B. Rawat, V. Chaudhary, R. Doku, Blockchain technology: emerging applications and use cases for secure and trustworthy smart systems, *J. Cybersec. Priv.* 1 (1) (2020) 4–18, <https://doi.org/10.3390/jcp1010002>.
- [14] A. Lampinen, V. Lehtinen, A. Lehmuskallio, et al., We're in it together: interpersonal management of disclosure in social network services, in: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ACM, 2011, pp. 3217–3226, <https://doi.org/10.1145/1978942.1979420>.
- [15] E. Alston, Blockchain and the law – legality, law-like characteristics, and legal applications, *The Economics of Blockchain and Cryptocurrency* (2022) 117–144, <https://doi.org/10.4337/9781800882348.00014>.
- [16] I. Chatziannakis, V. Liagkou, D. Salouros, et al., A security model for internet-based digital asset management systems, in: R. Morrison, D. Balasubramaniam, K. Falkner (Eds.), *Software Architecture*, Springer Berlin Heidelberg, Berlin, Heidelberg, 2008, pp. 326–329, https://doi.org/10.1007/978-3-540-88030-1_29.
- [17] H.R. Hasan, K. Salah, R. Jayaraman, et al., Blockchain-based solution for COVID-19 digital medical passports and immunity certificates, *IEEE Access* 8 (2020) 222093–222108, <https://doi.org/10.1109/ACCESS.2020.3043350>.
- [18] Alina Nastasache, Smart contracts: ¿qué son, cómo funcionan y qué aportan?, <https://academy.bit2me.com/que-son-los-smart-contracts/>, 2023. (Accessed 3 September 2024).
- [19] A. Castiglione, A.D. Santis, B. Masucci, et al., Hierarchical and shared access control, *IEEE Trans. Inf. Forensics Secur.* 11 (4) (2016) 850–865, <https://doi.org/10.1109/TIFS.2015.2512533>.
- [20] L. Marchesi, M. Marchesi, R. Tonelli, et al., A blockchain architecture for industrial applications, *Blockchain Res. Appl.* 3 (4) (2022) 100088, <https://doi.org/10.1016/j.bcr.2022.100088>.
- [21] C. Baliker, M. Baza, A. Alourani, et al., On the applications of blockchain in FinTech: advancements and opportunities, *IEEE Trans. Eng. Manag.* 71 (2024) 6338–6355, <https://doi.org/10.1109/TEM.2022.3231057>.
- [22] B. Cao, Z. Wang, L. Zhang, et al., Blockchain systems, technologies, and applications: a methodology perspective, *IEEE Commun. Surv. Tutor.* 25 (1) (2023) 353–385, <https://doi.org/10.1109/COMST.2022.3204702>.
- [23] IEEE, IEEE recommended practice for e-invoice business using blockchain technology, in: *IEEE Std 2142.1-2021*, 2021, pp. 1–18, <https://doi.org/10.1109/IEEESTD.2021.9381780>.
- [24] Z. Feng, H. Da, M. Xiao, et al., Supply chain finance in blockchain-based digital asset platform, in: *Proceedings of the 2022 4th International Conference on Blockchain Technology*, ACM, 2022, pp. 74–80, <https://doi.org/10.1145/3532640.3532650>.
- [25] T. Liu, J. Liu, J. Wang, et al., Anonymous storage and verification model of IIoT production status based on blockchain, in: *Proceedings of the 2021 4th International Conference on Blockchain Technology and Applications*, ACM, 2021, pp. 144–150, <https://doi.org/10.1145/3510487.3510508>.
- [26] R. Saraiva, A.A. Araújo, P. Soares, et al., MIRIAM: a blockchain-based web application for managing professional registrations of medical doctors in Brazil, in: *Proceedings of the 2021 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, IEEE, 2021, pp. 1–2, <https://doi.org/10.1109/ICBC51069.2021.9461051>.
- [27] A.-S. Kleinaki, P. Mytis-Gkometh, G. Drosatos, et al., A blockchain-based notarization service for biomedical knowledge retrieval, *Comput. Struct. Biotechnol. J.* 16 (2018) 288–297, <https://doi.org/10.1016/j.csbj.2018.08.002>.
- [28] X. Yu, Blockchain-based supply chain financial services using smart contract, in: *Proceedings of the 2021 4th International Conference on Blockchain Technology and Applications*, ACM, 2021, pp. 63–69, <https://doi.org/10.1145/3510487.3510497>.
- [29] E.J. Scheid, P. Kiechl, M. Franco, et al., Security and standardization of a notary-based blockchain interoperability API, in: *Proceedings of the 2021 Third International Conference on Blockchain Computing and Applications (BCCA)*, IEEE, 2021, pp. 42–48, <https://doi.org/10.1109/bcca53669.2021.9656961>.
- [30] P. Soares, R. Saraiva, I. Fernandes, et al., A blockchain-based customizable document registration service for third parties, in: *Proceedings of the 2022 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, IEEE, 2022, pp. 1–2, <https://doi.org/10.1109/ICBC54727.2022.9805500>.
- [31] M. Wöhrer, U. Zdun, Architectural design decisions for blockchain-based applications, in: *Proceedings of the 3rd IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, 2021, pp. 1–5, <https://doi.org/10.1109/ICBC51069.2021.9461109>.
- [32] V. Dwivedi, V. Pattanaik, V. Deval, et al., Legally enforceable smart-contract languages: a systematic literature review, *ACM Comput. Surv.* 54 (5) (2021) 1–34, <https://doi.org/10.1145/3453475>.
- [33] F. Spyghiger, C.J. Tessone, L. Zavolokina, et al., Incentivizing data quality in blockchain-based systems—the case of the digital cardossier, *Distrib. Ledger Technol.* 1 (1) (2022), <https://doi.org/10.1145/3538228>.
- [34] J. Guo, C. Li, G. Zhang, et al., Blockchain-enabled digital rights management for multimedia resources of online education, *Multimed. Tools Appl.* 79 (2020) 9735–9755, <https://doi.org/10.1007/s11042-019-08059-1>.

- [35] BOE Consolidated Legislation, Ley orgánica 3/2018, de 5 de diciembre, de protección de datos personales y garantía de los derechos digitales, <https://www.boe.es/buscar/pdf/2018/BOE-A-2018-16673-consolidado.pdf>, 2023. (Accessed 3 September 2024).
- [36] Government Gazette of the Parliament of Catalonia, Proyecto de ley sobre las voluntades digitales y de modificación de los libros segundo y cuarto del código civil de cataluña, <https://www.icab.es/es/actualidad/noticias/noticia/Proyecto-de-ley-sobre-las-voluntades-digitales-y-de-modificacion-de-los-libros-segundo-y-cuarto-del-Codigo-Civil-de-Cataluna/>, 2017. (Accessed 5 December 2023).
- [37] Regulation (UE) 2016/679 - European Parliament and the Council, Rgpd - normativa europea para la protección de datos personales, <https://www.boe.es/doue/2016/119/L00001-00088.pdf>, 2016. (Accessed 1 September 2024).
- [38] Web France Diplomacy, Política de confidencialidad y protección de la intimidad, <https://www.diplomatie.gouv.fr/es/navegacion/politica-de-confidencialidad-y-proteccion-de-la-intimidad/>, 2013. (Accessed 3 September 2024).
- [39] State of Connecticut, An act concerning access to decedent' electronic mail accounts, <https://www.cga.ct.gov/2005/act/Pa/2005PA-00136-R00SB-00262-PA.htm>, 2005. (Accessed 3 September 2024).
- [40] State of Delaware, Fiduciary access to digital assets and digital accounts-part v-fiduciary relations recovery of digital assets and digital accounts from a custodian, <https://delcode.delaware.gov/title12/c050/>, 2023. (Accessed 3 September 2024).
- [41] The New York State Senate, Legislative session, fiduciary access to digital assets and digital accounts-part v-fiduciary relations recovery of digital assets and digital accounts from a custodian, <https://www.nysenate.gov/legislation/bills/2013/A823>, 2013. (Accessed 3 September 2024).
- [42] Government of Spain, Ministerio de industria comercio y turismo, https://transparencia.gob.es/transparencia/gl/transparencia_Home/index/PublicidadActiva/OrganizacionYEmpleo/Funciones/Historico/Funciones-XII-Legislatura/Funciones-MINETAD.html, 2023. (Accessed 3 September 2024).
- [43] Parlamento Europeo, El fondo Europeo de desarrollo regional (FEDER), <https://www.europarl.europa.eu/factsheets/es/sheet/95/el-fondo-europeo-de-desarrollo-regional-feder->, 2023. (Accessed 3 September 2024).
- [44] BOE, Ley de propiedad intelectual, <https://www.boe.es/eli/es/rdlg/1996/04/12/1/con>, 2022. (Accessed 3 September 2024).
- [45] Bank of Spain, Normativa financiera, <https://entethalliance.org/membership/>, 2024. (Accessed 3 September 2024).
- [46] State Agency Official Gazette of the State, Ley 30/1991, de 20 de diciembre, de modificación del código civil en materia de testamentos, <https://www.boe.es/buscar/doc.php?id=BOE-A-1991-30534>, 1991. (Accessed 3 September 2024).
- [47] European Commission, Sucesiones y testamentos. Normas de la ue sobre sucesión internacional, https://commission.europa.eu/strategy-and-policy/policies/justice-and-fundamental-rights/civil-justice/family-law/successions-and-wills_es, 2024. (Accessed 3 September 2024).
- [48] Government of Spain, Derechos digitales ed.2023, <https://www.ontsi.es/es/publicaciones/Derechos-digitales-ed2023>, 2024. (Accessed 3 September 2024).
- [49] Bitcoin, Bitcoin tecnología, <https://bitcoin.org/es/>, 2023. (Accessed 3 September 2024).
- [50] ETH, Guía ethereum, <https://ethereum.org/es/>, 2023. (Accessed 3 September 2024).
- [51] bit2me Academy, ¿qué es ipfs?, <https://academy.bit2me.com/que-es-ipfs/>, 2023. (Accessed 3 September 2024).
- [52] Consumers and Users Organization, Cómo apuntarse a la Lista Robinson, <https://www.ocu.org/consumo-familia/derechos-consumidor/consejos/como-apuntarse-lista-robinson>, 2024. (Accessed 3 September 2024).
- [53] AWS, ¿Qué es una API RESTful?, <https://aws.amazon.com/es/what-is/restful-api/>, 2023. (Accessed 3 September 2024).
- [54] A. Khan, Graph analysis of the Ethereum blockchain data: a survey of datasets, methods, and future work, in: Proceedings of the 2022 IEEE International Conference on Blockchain (Blockchain), 2022, pp. 250–257, <https://doi.org/10.1109/Blockchain55522.2022.00042>.
- [55] Membership - Enterprise Ethereum Alliance, Enterprise Ethereum alliance, <https://entethalliance.org/>, 2013. (Accessed 3 September 2024).
- [56] Arrakis, Arrakis, <https://resources.arrakis.fi/arrakis-v2-protocol/introduction>, 2024. (Accessed 3 September 2024).
- [57] The Quorum Networks, <https://thequorumnetwork.com/>, 2024. (Accessed 3 September 2024).
- [58] Official Solidity Documentation, Smart contract factory, <https://docs.soliditylang.org/en/v0.8.0/contracts.html#contract-factories>, 2024. (Accessed 3 September 2024).
- [59] OpenAPI Specificatio, Version 2.0 introductions revision history definitions specification, <https://swagger.io/specification/v2/>, 2022. (Accessed 3 September 2024).
- [60] OAuth, ¿qué es oauth 2.0?, <https://auth0.com/es/intro-to-iam/what-is-oauth-2>, 2023. (Accessed 3 September 2024).
- [61] Solidity Org, Solidity is a tool that provides a structured and automated environment for testing smart contracts, <https://soliditylang.org/>, 2024. (Accessed 3 September 2024).
- [62] Docker, Docker community edition (ce) is the free and open source version of docker, <https://www.docker.com/>, 2024. (Accessed 3 September 2024).
- [63] MicroFocus, Loadrunner is a microfocus framework designed to perform performance and load testing in applications, allowing you to simulate real conditions of use and evaluate the behavior of the system under different scenarios, <https://www.opentext.com/products/performance-engineering>, 2024. (Accessed 3 September 2024).
- [64] icommunity, How does decentralized digital identity work, <https://icommunity.io/que-es-la-identidad-digital-descentralizada/>, 2023. (Accessed 3 September 2024).